METHOD 26 / A COMPLETE BEGINNER GUIDE

Hayashi-Yoshida covariance

How do you estimate covariance when FX quotes arrive at different times? The Hayashi-Yoshida estimator sums products of returns on overlapping intervals only.

The output is a **covariance**, not a bounded correlation. Daily synced closes are the wrong input when your timestamps are async. Teaching example only; experiments not started.

Read it in three parts

First: Overlaps and FX percent returns. **Next:** Hayashi and Yoshida (2005) and the main sum. **Last:** Hand calculation, limits, and link to folder 25.

Pages	What you will learn
2	Idea; covariance vs correlation
3	FX 1.1000 to 1.1110
4	Intervals and overlap indicator
5	Hayashi & Yoshida 2005
6-9	Formula and hand sum
10-11	Why synced daily closes fail
12-14	Limits, software, sources

Prepared for the folder: 26 - Hayashi-Yoshida Covariance
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02 / THE BASIC IDEA

What does HY covariance do?

```
C_HY = SUM_i SUM_j dX_i * dY_j * 1( I_i overlaps J_j )
```

dX_i is the return on X over its own observation interval I_i. dY_j is the return on Y over J_j. The indicator is 1 when the time ranges overlap, else 0.

Covariance, not correlation

C_HY is on the scale of return products (for example decimal times decimal). It is **not** forced into [-1, +1]. To get a correlation shape you must divide by estimated variances with the same overlap logic.

Is C_HY a percent?

No. Returns may be labelled in percent; covariance inherits return units squared in the usual variance scaling story.

03 / RETURNS IN PLAIN NUMBERS

FX percent math on one move

```
Simple return on a leg: 1.1110/1.1000 - 1 = 0.01
Log return: ln(1.1110/1.1000) = about 0.009950
```

HY uses returns tied to each quote interval, not one artificial daily pair. Build dX_i from the price change between the tick times that define I_i.

Document whether you use log or simple returns before you sum products.

04 / DATA RULES

Intervals and overlaps

Record timestamps

Each interval needs start and end in a common clock. Overlap means the ranges intersect with positive length; touching at one instant may be excluded depending on implementation—match your library to [1].

Async is normal in FX

EUR may update on one bank stream while JPY updates on another. HY avoids pretending both moved on a fictional shared close.

Versus folder 25 realized correlation

Realized rho with aligned intraday buckets is a different object. HY is for mismatched interval grids.

Research notes

Hayashi, T., & Yoshida, N. (2005) [1]

Bernoulli, 11(4), 729-749. Gives the overlap-sum covariance estimator for non-synchronously observed diffusions and studies its properties.

Folder 25 cites the same paper for realized and async context. This folder focuses on the covariance sum itself.

Claims in this guide

Teaching overlap arithmetic only. No invented tick covariance for live pairs until experiments exist.

06 / DICTIONARY

Meet the symbols

Symbol	Meaning
I_i	Time range for the i-th return on X.
J_j	Time range for the j-th return on Y.
dX_i, dY_j	Returns on those intervals.
1(overlap)	1 if I_i and J_j overlap, else 0.
C_HY	Hayashi-Yoshida covariance estimate.

Some codes use half-open intervals [t_start, t_end). State your convention in experiment logs.

07 / WHAT EACH PART DOES

The main formula

```
C_HY = SUM_i SUM_j dX_i * dY_j * 1( I_i intersects J_j )
```

Double sum

Every X return pairs with every Y return, but zero weight when intervals do not overlap. Overlapping segments contribute their product once per qualifying pair.

Scaling to correlation (optional)

A HY-type correlation divides C_HY by sqrt(HY variance estimates for X and Y). That extra step is not automatic in the covariance formula above.

Hand setup: intervals

Two X intervals and two Y intervals on a unit timeline (hours as labels):

Leg	Interval	Return
X	I1: [0, 1]	dX1 = 0.010
X	I2: [1, 3]	dX2 = 0.005
Y	J1: [0, 2]	dY1 = 0.012
Y	J2: [2, 3]	dY2 = 0.008

Overlap grid (1 = yes): I1 with J1 yes, I1 with J2 no, I2 with J1 yes, I2 with J2 yes.

09 / WORKED EXAMPLE PART TWO

Hand finish: the sum

```
Pairs counted:
I1-J1: 0.010 * 0.012 = 0.00012
I1-J2: skipped (no overlap)
I2-J1: 0.005 * 0.012 = 0.00006
I2-J2: 0.005 * 0.008 = 0.00004

C_HY = 0.00012 + 0.00006 + 0.00004 = 0.00022
```

Compare to the naive product of daily totals: you would miss which pieces of time actually shared information.

Sign matters

Negative dX_i or dY_j flip the product sign. Covariance can be negative without being invalid.

10 / INPUT MISTAKE

Why daily synced closes are wrong

The error

Taking one close per day for X and one for Y forces a single interval pair. That discards timing inside the day and misstates overlap when quotes are async.

Better habit

Store tick or bar timestamps, build I_i and J_j from actual updates, then apply the indicator sum.

If you only have daily closes, HY is not the right tool unless your question truly is daily and synchronous.

Limits

Not a trading rule

Covariance magnitude does not include costs, causality, or stability.

Microstructure noise

Very noisy ticks inflate products; pre-averaging or kernel methods may be needed in production code.

Not bounded

Do not report C_HY as if it were a correlation without explicit normalisation.

Software sketch

```
dx = [0.010, 0.005]
dy = [0.012, 0.008]
overlap = [[1,0],[1,1]]
c = 0.0
for i, xi in enumerate(dx):
for j, yj in enumerate(dy):
if overlap[i][j]:
c += xi * yj
print(c) # 0.00022
```

Production HY code builds overlap from timestamps automatically. Verify against one hand grid before scaling up.

Sources and further reading

[1] Hayashi, T., & Yoshida, N. (2005). On covariance estimation of non-synchronously observed diffusion processes. Bernoulli, 11(4), 729-749.
Primary HY covariance reference.
Open source

[2] Folder 25 Volatility and Realized Correlation.
Aligned-bucket realized versus async HY.
Open source

14 / QUICK QUESTIONS

Check your understanding

Is C_HY a correlation? No, unless you normalise separately. **Can I use two daily closes?** Not for async HY as taught here. **Does overlap matter?** Yes; the indicator is the core of the estimator.

When experiments start, save the overlap matrix for one day and reconcile it to the double sum in your HY library.

Workflow before experiments

1. State the sampling reality

Async ticks, unequal bar grids, or true daily sync—pick the estimator to match.

2. Unit-check returns

Include the 1.1000 to 1.1110 move as a decimal return sanity row in experiment notes.

3. Log interval rules

Half-open versus closed intervals change edge overlaps. Copy the rule from your package docs.

4. Report covariance with units

Say decimal log returns, sample window, and whether you also publish a HY correlation variant.

Experiments not started. No live pair HY numbers in this folder until v001 with snapshot id.